

BUSHY PARK, AustraliaWMO: **949640**Lat: **42.7097S**Lon: **146.8983E**Elev: **27**StdP: **101.00**Time Zone: **10.00 (AUS)**Period: **98-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-1.7	-0.7	-0.9	3.5	8.1	-0.2	3.7	5.6	11.3	12.7	10.1	11.8	0.9	330	0.550

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	30.0	17.7	27.6	17.1	25.4	16.2	19.2	26.2	18.2	24.7	17.3	23.4	5.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	12.3	20.5	16.0	11.4	20.0	14.8	10.5	19.4	54.7	26.0	51.6	24.8	48.6	23.4	26.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	8.9	7.7	-3.5	35.3	1.0	1.9	-4.2	36.7	-4.8	37.8	-5.3	38.9	-6.1	40.3		
(5)			-0.9	21.2	1.0	0.8	-1.6	21.7	-2.2	22.2	-2.7	22.7	-3.4	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	17.2	16.8	15.0	11.7	9.3	6.9	6.7	7.9	9.8	11.5	13.7	15.4
	DBStd	4.57	3.13	3.10	3.07	2.72	2.73	2.75	2.43	2.31	2.32	2.81	2.96	3.10
	HDD10.0	390	0	0	2	13	46	98	105	74	32	16	3	1
	HDD18.3	2453	60	61	112	200	281	343	360	324	257	213	142	100
	CDD10.0	1043	223	191	155	64	24	6	3	7	25	62	115	169
	CDD18.3	64	24	18	8	1	0	0	0	0	0	1	3	10
	CDH23.3	979	355	238	101	6	0	0	0	0	1	19	82	177
	CDH26.7	292	123	69	22	1	0	0	0	0	0	3	19	54
Wind	WSAvg	3.1	3.7	3.4	3.0	2.5	2.6	2.2	2.4	2.9	3.4	3.5	3.5	3.7
Precipitation	PrecAvg	591	37	34	40	49	47	44	58	62	59	57	54	47
	PrecMax	879	93	130	95	157	114	93	148	127	153	182	129	141
	PrecMin	357	6	2	8	5	8	13	17	15	17	19	4	2
	PrecStd	105	20	27	22	29	30	23	33	30	31	29	26	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	32.5	29.5	24.1	19.6	16.5	15.8	18.1	21.1	25.8	29.3	31.7
		MCWB	18.5	18.9	17.7	15.4	13.1	12.2	10.5	11.6	12.6	14.6	16.7	17.3
	2%	DB	30.0	29.0	26.0	21.2	17.3	14.7	14.0	15.7	18.4	22.3	25.6	27.7
		MCWB	17.9	18.0	16.7	14.3	12.2	10.6	9.6	10.5	11.8	14.1	15.8	16.9
	5%	DB	27.2	26.5	23.6	19.2	15.7	13.3	12.6	14.2	16.7	19.9	22.8	24.8
		MCWB	16.9	16.8	15.9	13.4	11.3	9.9	9.1	9.8	11.0	12.9	14.8	16.0
	10%	DB	24.6	24.1	21.5	17.4	14.4	12.0	11.4	12.9	15.1	17.7	20.5	22.6
		MCWB	15.9	16.1	15.0	12.5	10.6	9.1	8.4	9.0	10.2	12.0	13.9	14.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	20.9	19.4	16.6	14.1	13.4	11.2	12.4	13.8	16.9	18.9	19.4
		MCDB	29.1	26.9	25.8	21.4	18.0	15.1	14.0	15.8	19.0	21.5	24.0	26.7
	2%	WB	19.0	19.2	17.9	15.1	12.8	11.2	10.2	11.2	12.5	15.1	17.0	18.1
		MCDB	26.4	26.2	23.4	19.3	16.1	14.0	13.3	14.9	16.8	20.1	23.6	24.2
	5%	WB	17.8	18.1	16.8	14.0	11.9	10.2	9.4	10.2	11.6	13.6	15.7	16.8
		MCDB	24.7	24.6	22.1	18.1	14.9	12.9	12.4	13.7	15.8	18.7	21.4	23.4
	10%	WB	16.6	16.9	15.7	13.0	11.0	9.4	8.7	9.4	10.6	12.3	14.3	15.6
		MCDB	23.2	22.5	20.7	16.8	14.0	11.9	11.3	12.6	14.8	17.3	19.6	21.6
Mean Daily Temperature Range		MDBR	13.5	13.1	12.2	11.8	9.6	8.9	9.1	10.1	10.8	11.5	12.0	12.5
	5% DB	MCDBR	19.4	17.9	16.3	14.9	11.3	10.3	10.4	12.0	13.5	15.9	17.3	17.7
		MCWBR	7.9	7.6	7.7	8.2	6.7	6.4	6.3	6.7	7.4	8.2	8.4	7.8
	5% WB	MCDBR	16.1	14.7	14.3	13.4	10.7	9.6	9.2	11.1	12.3	14.3	15.4	15.3
		MCWBR	7.3	7.1	7.3	7.9	6.9	6.5	6.2	6.7	7.3	8.1	8.3	7.5
Clear-Sky Solar Irradiance	taub	0.350	0.336	0.339	0.317	0.308	0.294	0.294	0.310	0.336	0.341	0.339	0.343	
	taud	2.476	2.533	2.512	2.559	2.565	2.588	2.590	2.548	2.457	2.456	2.486	2.488	
	Ebn at Noon	975	962	911	862	797	777	807	858	898	946	980	989	
	Edh at Noon	114	102	95	77	64	57	61	76	98	109	112	114	
All-Sky Solar Radiation	RadAvg	6.46	5.55	4.11	2.78	1.69	1.34	1.53	2.30	3.43	4.84	5.84	6.40	
	RadStd	0.51	0.33	0.24	0.17	0.12	0.11	0.08	0.16	0.19	0.29	0.46	0.40	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	+0.32	N/A	N/A	+0.66	N/A	N/A	N/A	-95	+103	+12

Nomenclature: See separate page